Analytics Capstone Project (ANCP)

Sparta: Wardrobe Demand Analytics for Brands

Deliverable D3: EDA notebook and two-page insight summary

Group number	M
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Files	ANCP_TM_D3_eda.ipynb, ANCP_TM_D3_eda.html, ANCP_TM_D3_summary.pdf

Sparta: what the purchase data says

ANCP D3 · EDA insight summary · Group M · Linked outcomes CO2, CO4 · Data: H&M 10% customer sample (3,205,663 rows, Sep 2018 to Sep 2020) as a proxy for purchase behaviour while the Sparta pilot has no data

> **Bottom line.** The data support building Sparta's outfit suggestions to *complete* outfits and its vendor insights to expect fast-changing demand, not a taste model. Fewer than half of garment visits (45%) contain a complete outfit, but baskets are not thin on average (30% single-unit visits; 3.5 units per visit). History helps mainly through exact re-buys, and demand turns over within weeks. The 10% sample matches the full file on every representativeness check, so the main risks are the proxy (European, no gender or store field) and the 2020 break, not sampling.

The five insights that matter most, and how each shapes modelling

Insight	Evidence	Shapes modelling
1 The gap is the incomplete outfit, not the one-item basket.	Garment visits: complete 45%, top only 38%, bottom only 17%. Single-unit visits 30% (95% CI 30% to 30%); thin baskets sit in one channel (38% vs 25%; meaning of the code unverified).	Aim outfit suggestions (FR-52) at wardrobes with a top but no bottom, or the reverse; expect baskets of about 3.5 units, not 1; revise uplift assumptions in D5.
2 History predicts re-buying, not taste.	Exact-article repeat: 5.4% vs 0.3% for a same-week random draw (19×). Product type 1.26×, colour 1.07×.	Start with repeat and recency features; add taste features only if they beat that. No taste model in the pilot (BR-28).
3 Demand turns over within weeks.	Last week's top 12 covers 1.6% of next week's units, top 100 covers 8.4%; top-100 overlap 0.29 at 1 week (noise ceiling 0.37), 0.13 at 4 weeks. 39% of weekly units come from articles first sold up to 8 weeks earlier.	Rebuild the most-popular baseline weekly; validate by date; give new articles an attribute-based fallback; add season terms to the forecast.
4 Heavy buyers dominate; many shoppers have no history; older shoppers buy least per visit.	Top 10% of customers buy 48% of units; 32% have one visit. Single-unit share by age: 65+ 43% vs 25 to 34 27%.	Split by customer, report by activity tier and age band, weight heavy buyers, and keep a popularity or rule-based fallback for new users.
5 Co-purchase cannot identify complements; colour is mostly neutrals.	Median adjusted lift: top×bottom 1.35, top×top 1.36 (no difference). Black, White, Blue = 62% of units; light share 42% in summer vs 30% in winter.	Use co-purchase only to suggest candidate pairs to the style specialist; outfit quality stays in rules R1 to R8. The colour rule (R3) will rarely bind.

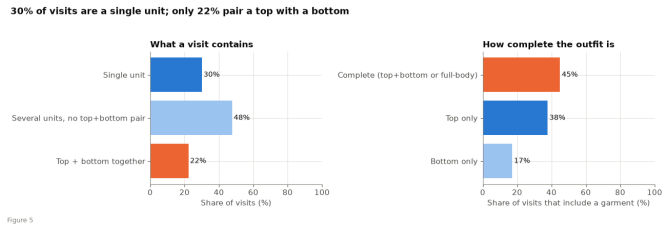


Figure 5. Most visits are not single-unit; outfits are incomplete.

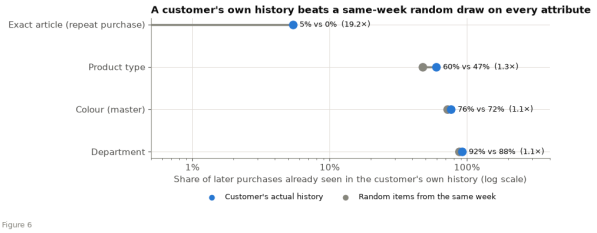


Figure 6. History is strongest for exact re-buys.

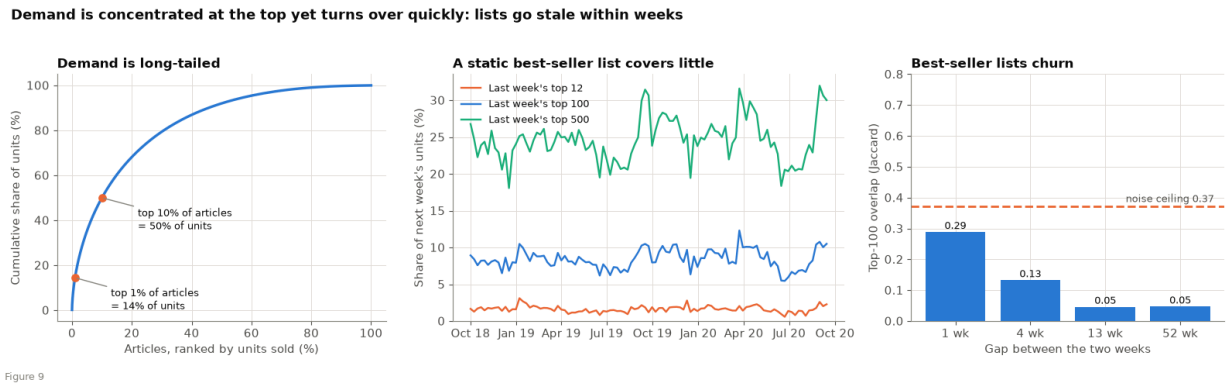


Figure 9. Demand is concentrated at the top and the top list changes quickly. Figures are from ANCP_TM_D3_eda.ipynb.

Seven hypotheses tested against rules fixed before loading the data

ID	Question	Hypothesis	Evidence vs rule	Verdict
H1	BQ1	Thin baskets, outfit rarely complete	Single-unit 30% (rule ≥ 50%); top+bottom 22% (rule ≤ 15%)	Not supported
H2	BQ2	History predicts next purchase (≥ 1.5×)	Type 1.26×, colour 1.07×; exact article 19× (not in rule)	Not supported
H3	BQ3	Top–bottom pairs are complementary	Median lift 1.35 (rule ≥ 1.2); top×top 1.36, so not specific to complements	Supported
H4	BQ4	Long tail, fast churn	Top 1% of articles hold 14% (rule < 25%); overlap 0.29 at 1 wk (ceiling 0.37), 0.13 at 4 wk	Supported
H5	BQ5	Segments differ	Cramér's V 0.06 (rule ≥ 0.15) fails; single-unit gap 17 pts across age bands passes	Not supported
H6	BQ2/5	Heavy buyers dominate	Top 10% of customers buy 48% of units (rule ≥ 40%); Gini 0.64	Supported
H7	BQ3/4	Colour concentrated and seasonal	Top 3 colours 62% (rule ≥ 50%); light share 42% vs 30%	Supported

Four supported, three not. The three failures each correct an assumption we started with (thin baskets, taste-based history, age-driven department mix).

Representativeness and bias, repeated on raw and cleaned data

Sample vs full population: 6 of 6 checks pass on raw data and the cleaned frame: department mix (largest gap 0.05 pp), monthly profile (0.06 pp), channel mix (0.02 pp), basket size, purchases per customer (23.5 vs 23.3), customer age and club status (largest |SMD| 0.02). Cleaning (D2 report, Table 3): duplicate rows kept as quantity (296,434 rows, units preserved); Baby/Children (3.8% of units), non-apparel and under-18 customers excluded; 3,066,344 units in scope; Menswear is 5.8% of them. Remaining bias: missing age, FN flag or club status mark light buyers (median 1 to 2 visits vs 3 to 4); units in Mar–Aug 2020 were −13% vs 2019 and channel-2 share rose from 70% to 75%; no gender, size or store field; European assortment; price is normalised.

Insight log (full table with evidence and so-what in the notebook, section 9)

ID	Insight	Decision it informs	Confidence
I1	Baskets are not thin on average; outfits are incomplete.	Outfit-suggestion scope (FR-52); basket-size baseline	High (H&M data); Medium for transfer to Sparta
I2	Thin baskets sit in one sales channel.	Channel meaning to confirm before use	Medium (channel label unverified)
I3	A customer's own history is a strong predictor only at the level of the exact article.	Ranking features	Medium (ratio depends on history length; H&M only)
I4	Purchases show which types are bought together, but not which are complements.	Outfit rules R1 to R8 (candidates only)	Medium
I5	Demand turns over fast.	Baseline, forecast, validation design	High (direction); Medium (levels, 10% sample is noisy)
I6	A minority of heavy buyers dominates the data and a large group has almost no history.	Splits, weighting, cold start	High
I7	Segments differ more in basket thinness than in what they buy.	Segment reporting (D5)	High
I8	Colour choice is concentrated in three neutrals and follows the season.	Colour rule R3; seasonality	Medium (colour attribute is the catalogue's, not the shopper's)
I9	The sample is not the bias; the proxy and the period are.	Validation windows; transfer to the Sparta pilot	High (sample); Medium (cause of the 2020 break)
I10	The "single-item habit" group (our proxy for the "nothing to wear" user) is visible but small; transactions cannot show confidence or effort.	Business-case sizing; pilot data	Low

Confidence: High = large sample, narrow interval, same direction on raw and cleaned data; Medium = depends on an unverified label, noisy measure or transfer to Sparta; Low = behavioural proxy.

Limits and next steps

H&M customers are not Sparta's pilot users; every number is a pattern to re-validate on pilot data. The "nothing to wear" user cannot be seen in transactions; our proxy (4+ visits, 60%+ single-unit) covers 10% of repeat customers and 3.5% of their units, and pilot data must confirm it. Next: confirm what the sales-channel code means; carry the weekly baseline, date-and-customer splits and age-band checks into D4 and D5. Code and files: https://github.com/cprashanth666666-sys/Captain/tree/D3